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Project Title: Personal Finance Tracker

Problems Being Solved:

This program helps manage personal finances because for some it can be difficult to keep track of all their financial information such as income, expenses, and spending. Overspending and managing a monthly budget is a problem that many deal with. The problem this project solves is to understand how to be better with their money. Demonstrating to the user how a program can help them track their budget, record income and expenses.

Solution Implemented in the Project:

This program provides a menu based system that gives you multiple options and allows the user to manage their financial information. It allows the user to enter their monthly budget, add income to their total income, add expenses and subtract from the total, view a summary that includes all their financial information, and save this information onto a text file. The program uses variables, lists, functions, classes, objects, exception handling, file operations, and a custom module that organizes and processes data.

Algorithm Implementation Explanation:

The program starts off with asking the user to enter their monthly budget and then storing that information. It then creates a Budget object from the class. Then the menu will show prompting the user to enter one of the options. The menu uses conditional statements to determine which operation the user wants to perform and the program will only end once the user chooses to exit. If the user chooses to add income, the program will take the income amount and add it to the total income. If the user chooses to add expense, the program will take the expense amount, category, and description. This information is stored into a list and an object in order to organize all the data. The program calculates the remaining money by subtracting the total expenses from the total income and the custom module calculates the remaining balance and checks whether or not the user is within their budget. If the user wants to open a file and keep this information in there, the program will use file operations to write the summary into a text file.

Program Objectives:

The objectives are to allow users to create and manage a monthly budget, allow users to input their income and expenses, find how much money remains after expenses, help the user understand how to save better, save the financial information, and demonstrate the Python concepts from this course.

User Interaction and Purpose:

The user interacts through a menu based system where the user can enter budget amount, income, expenses, categories, and descriptions. The purpose of this program is to provide the user with an easier, more straightforward way of organizing financial information and making better decisions with their money.

Program Limitations:

A limitation the program has is that once the program has been closed, the information that has been entered will be lost and have to be entered once again. Another limitation is that there are no graphs or models that show advanced analytics. The program also only supports one user's financial information.

Recommendations for Improvement:

An improvement to this program is to have a database that stores financial records so that the user doesn't have to enter their information every time they open the program. Also, adding a chart that shows the user's spending habits could also help the user visualize how they are handling their money.

BEGIN

```
IMPORT finance_tools module

CREATE Transaction class
CREATE Budget class
CREATE Expense subclass from Transaction
```

```
SET budget, income, expenses to 0
```

```
CREATE expense list
```

```
INPUT monthly budget
```

```
IF budget input is invalid
    SET budget to 0
END IF
```

```
DEFINE add_income function
    INPUT income amount
    ADD amount to income total
END FUNCTION
```

```
DEFINE add_expense function
    INPUT expense amount

    IF expense is negative
        DISPLAY error message
    ELSE
        ADD expense to expense total
        STORE expense information
        CREATE Expense object
    END IF
END FUNCTION
```

```
DEFINE display_summary function
    CALCULATE remaining money
    DISPLAY budget, income, expenses, and
    remaining money
    CHECK budget status
END FUNCTION
```

```
DEFINE save_file function
    OPEN summary file
    WRITE financial information
```

```
CLOSE file
END FUNCTION
```

```
SET choice = 0
```

```
WHILE choice is not 5
```

```
    DISPLAY menu options
```

```
    INPUT user choice
```

```
    IF choice = 1
        CALL add_income
```

```
    ELSE IF choice = 2
        CALL add_expense
```

```
    ELSE IF choice = 3
        CALL display_summary
```

```
    ELSE IF choice = 4
        CALL save_file
```

```
    ELSE IF choice = 5
        DISPLAY exit message
```

```
    ELSE
        DISPLAY invalid choice
```

```
    END IF
```

```
END WHILE
```

```
BEGIN finance_tools module
```

```
FUNCTION calculate_remaining
    RETURN income - expenses
END FUNCTION
```

```
FUNCTION check_budget
    IF expenses > budget
        RETURN over budget message
    ELSE
        RETURN within budget message
    END IF
END FUNCTION
END
```